

```
root@node-0-0:~# sudo apt-get update
sudo apt-get install -y libpam-google-authenticator
Hit:1 http://security.debian.org/debian-security bookworm-security InRelease
Hit:2 http://deb.debian.org/debian bookworm InRelease
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  libpam-google-authenticator
0 upgraded, 1 newly installed, 0 to remove and 216 not upgraded.
Need to get 45.5 kB of archives.
After this operation, 138 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main amd64 libpam-google-authenticat
or amd64 20191231-2 [45.5 kB]
Fetched 45.5 kB in 0s (136 kB/s)
Selecting previously unselected package libpam-google-authenticator.
(Reading database ... 151453 files and directories currently installed.)
Preparing to unpack .../libpam-google-authenticator_20191231-2_amd64.deb ...
Unpacking libpam-google-authenticator (20191231-2) ...
Setting up libpam-google-authenticator (20191231-2) ...
Processing triggers for man-db (2.11.2-2) ...
root@node-0-0:~#
```

```
root@node-0-0:~# google-authenticator
```

Do you want authentication tokens to be time-based (y/n) y

Warning: pasting the following URL into your browser exposes the OTP secret to Google:

<https://www.google.com/chart?chs=200x200&chld=M|0&cht=qr&chl=otpauth://totp/root@node-0-0%3Fsecret%3>



Code incorrect (correct code 302486). Try again.

Enter code from app (-1 to skip): 097720

Code confirmed

Your emergency scratch codes are:

40481008

86929243

17199169

65512815

93306075

Do you want me to update your `"/root/.google_authenticator"` file? (y/n) █

Do you want me to update your `"/root/.google_authenticator"` file? (y/n) y

Do you want to disallow multiple uses of the same authentication token? This restricts you to one login about every 30s, but it increases your chances to **notice** or even prevent man-in-the-middle attacks (y/n) y

By default, a new token is generated every 30 seconds by the mobile app. In order to compensate for possible time-skew between the client and the server, we allow an extra token before and after the current time. This allows for a time skew of up to 30 seconds between authentication server and client. If you experience problems with poor time synchronization, you can increase the window from its default size of 3 permitted codes (one previous code, the current code, the next code) to 17 permitted codes (the 8 previous codes, the current code, and the 8 next codes). This will permit for a time skew of up to 4 minutes between client and server.

Do you want to do so? (y/n) n

If the computer that you are logging into isn't hardened against brute-force login attempts, you can enable rate-limiting for the authentication module.

By default, this limits attackers to no more than 3 login attempts every 30s.

Do you want to enable rate-limiting? (y/n) y

root@node-0-0:~# █

```
GNU nano 7.2 /etc/pam.d/login *
#
# The PAM configuration file for the Shadow 'login' service
#
auth required pam_google_authenticator.so
# Enforce a minimal delay in case of failure (in microseconds).
# (Replaces the 'FAIL_DELAY' setting from login.defs)
# Note that other modules may require another minimal delay. (for example,
# to disable any delay, you should add the nodelay option to pam_unix)
auth optional pam_faildelay.so delay=3000000

# Outputs an issue file prior to each login prompt (Replaces the
# ISSUE_FILE option from login.defs). Uncomment for use
# auth required pam_issue.so issue=/etc/issue

# Disallows other than root logins when /etc/nologin exists
# (Replaces the 'NOLOGINS_FILE' option from login.defs)
auth requisite pam_nologin.so

# SELinux needs to be the first session rule. This ensures that any
# lingering context has been cleared. Without this it is possible
# that a module could execute code in the wrong domain.
# When the module is present, "required" would be sufficient (When SELinux
# is disabled, this returns success.)
session [success=ok ignore=ignore module_unknown=ignore default=bad] pam_selinux.so close

# Sets the loginuid process attribute
session required pam_loginuid.so

# Prints the message of the day upon successful login.
# (Replaces the 'MOTD_FILE' option in login.defs)
# This includes a dynamically generated part from /run/motd.dynamic
# and a static (admin-editable) part from /etc/motd.
session optional pam_motd.so motd=/run/motd.dynamic
session optional pam_motd.so noupdate

# SELinux needs to intervene at login time to ensure that the process
# starts in the proper default security context. Only sessions which are
# intended to run in the user's context should be run after this.

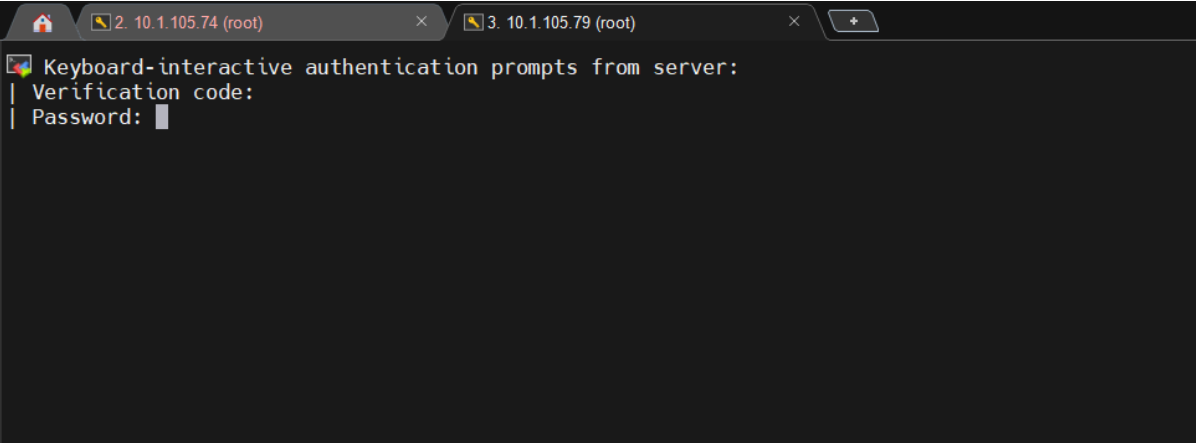
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^M Undo       ^_ Set Mark   ^= To Bracket
^X Exit      ^R Read File  ^R Replace    ^U Paste      ^I Justify    ^_ Go To Line  ^- Redo       ^- Copy      ^_ Where Was
```

```
# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes
```



The image shows a terminal window with two tabs. The first tab is titled '2. 10.1.105.74 (root)' and the second tab is titled '3. 10.1.105.79 (root)'. The terminal content shows a prompt for keyboard-interactive authentication from a server, with fields for 'Verification code:' and 'Password:'.

```
Keyboard-interactive authentication prompts from server:
| Verification code:
| Password: █
```